

Inference Rules

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THE INFERENCE RULES OF CHAPTER II

For Chapter II, we add no new methods for Boxing & Canceling or for making Assumptions. We add only inference rules. For each of the three new sentential connectives, “ $\wedge$ ”, “ $\vee$ ”, and “ $\leftrightarrow$ ”, there are new, primitive inference rules that all us to *Introduce* a formula with that as its main connective, and there are also new inference rules that allow us to *Eliminate* (i.e. make use of) a formula with that as its main connective.

The primitive Introduction and Elimination rules for each of the new connectives are as follows.

PRIMITIVE RULES

INTRODUCTION RULES

Adjunction ADJ	Addition ADD ADDl ADDr		Conditional Biconditional CB
$\frac{P}{Q} \quad \frac{Q}{P \wedge Q}$	$\frac{P}{Q \vee P}$	$\frac{P}{P \vee Q}$	$\frac{P \rightarrow Q}{Q \rightarrow P}$ $\frac{P \leftrightarrow Q}{P \leftrightarrow Q}$
Order of premises doesn't matter.	Add on the left	Add on the right	Order of premises doesn't matter.

ELIMINATION RULES

Simplification S Sl Sr	Modul Tollendo Ponens MTP MTPl MTPr		Biconditional Conditional BC BCl BCr	
$\frac{P \wedge Q}{P}$ $\frac{P \wedge Q}{Q}$	$\frac{P \vee Q}{\sim P}$ $\frac{\sim P}{Q}$	$\frac{P \vee Q}{\sim Q}$ $\frac{\sim Q}{P}$	$\frac{P \leftrightarrow Q}{P \rightarrow Q}$	$\frac{P \leftrightarrow Q}{Q \rightarrow P}$
S on the left S on the right	MTP on the left	MTP on the right	BC on the left	BC on the right

DERIVED RULES

In Chapter II, we also allow *Derived Rules* of inference. Derived rules are based on theorems. In order to enable a derived rule, you must prove the underlying theorem. Both the Problem List, accessed through the Select button, and the Available Rules window, accessed through the Rules button, indicate which theorems must be proved to activate each of the derived rules.

The most important derived rules are those associated with specific strategies for completing derivations (see separate “Advice” document). These include:

Conditional Disjunction	DeMorgan Laws	DeMorgan Laws
CDJ (Deriv 2.104 T45)	DM (Deriv 2.119 T65)	DM (Deriv 2.120 T66)
$\frac{\sim P \rightarrow Q}{P \vee Q}$	$\frac{\sim(P \wedge Q)}{\sim P \vee \sim Q}$	$\frac{\sim(P \vee Q)}{\sim P \wedge \sim Q}$
Negation of a Conditional	Negation of a Biconditional	Negation of a Biconditional
NC (Deriv 2.101 T40)	NB (Deriv 2.126 T90)	RT87 (Deriv 2.087 T87)
$\frac{\sim(P \rightarrow Q)}{P \wedge \sim Q}$	$\frac{\sim(P \leftrightarrow Q)}{P \leftrightarrow \sim Q}$	$\frac{\sim(P \leftrightarrow Q)}{(P \wedge \sim Q) \vee (\sim P \wedge Q)}$

Do not neglect the other specified derived rules, all of which will prove valuable in specific situations.

THEOREMS

In addition to the named derived rules, any instance of a theorem you have already proved can be used in a derivation. The justification simply consists of the theorem name, e.g. “T20”.

THEOREMS AS RULES

Furthermore, if the theorem is a conditional or biconditional, it can be used as a rule simply by prefixing “R” to the theorem name. For example, once theorem T20: $(P \rightarrow \sim P) \rightarrow \sim P$, is proved, the corresponding rule RT20: $P \rightarrow \sim P \therefore \sim P$, is automatically enabled.

It is often considerably more convenient to use the corresponding rule rather than a theorem. Use of a theorem requires the user to provide the scheme of substitution manually, but when a rule is used, the program can usually figure out the substitution on its own.

Biconditional theorems yield two forms of the corresponding rule. Thus theorem T115: $(P \rightarrow \sim P) \leftrightarrow \sim P$, yields two forms for the rule RT115: $(P \rightarrow \sim P) \therefore \sim P$, and $\sim P \therefore (P \rightarrow \sim P)$. Since the two sides of most biconditional theorems have different forms, usually only one form of the rule will be applicable (and the program can figure this out). However, for those cases in which the user wishes to specify the form used, the suffixes “L” and “R” indicate, respectively, the left to right form, and the right to left form. Thus “RT115L” refers only to: $(P \rightarrow \sim P) \therefore \sim P$, and “RT115R” refers only to: $\sim P \therefore (P \rightarrow \sim P)$.

If the rule corresponding to a theorem has a conjunction as a premise, an alternative form of the rule containing each of the conjuncts as separate premises is also enabled. Thus, theorem 88:

$P \wedge \sim Q \rightarrow \sim(P \leftrightarrow Q)$ yields two forms for the rule RT88: $(P \wedge \sim Q) \therefore \sim(P \leftrightarrow Q)$, and $P, \sim Q \therefore \sim(P \leftrightarrow Q)$. The latter form avoids having to first apply ADJ.

USER RULES

You may find that certain patterns of inference that take several steps to justify are frequently repeated, and you may therefore wish to have a single rule as a shortcut. However, you may not find the inference pattern you wish to use among the derived rules or the theorems. In this case, you can create your own User Rule. The steps are simple:

1. Open the User Problem screen by clicking the User button.
2. Use the Toolbox (right click) or keyboard to fill in the problem in a way to mirror the inference pattern of the rule you wish to use, for example, “ $\sim P \rightarrow \sim Q. Q \therefore P$ ” is a handy variant of Modus Tollens. Don’t include the quotes, and remember to add a period, “.”, after each premise but the last, and a Therefore-sign, “ $\therefore$ ” between the premises and the conclusion. (On the keyboard, Ctrl+Shift+T inserts the Therefore-sign.)
3. Give a derivation for the problem, and check it to make sure it is correct.
4. Save the problem with a name beginning with the two letters “UR” (case doesn’t matter), for example, “urMTDN”. You can use any letter or numeral or any character on the top row of keys (! @ # \$ % ^ & * () _ + - =), but no spaces.
5. The new User Rule should now appear at the top of the Rules window, and can be used by citing the name you gave it (including the mandatory “UR” prefix).
6. User Problems are stored at the bottom of the Select list. To delete a User Rule, select and load the User Problem with that name, then delete it by clicking the Delete button (at the bottom of the screen). When queried, choose, “Delete this problem?”.
7. There is at present no way to rename a User Rule other than by creating a new rule and deleting the old, so choose your User Rule names carefully.

In proving theorems in order to use them or to use derived rules based on them, it is *not* required that the theorems be proved in any particular order. At any stage, any theorems that are proved are usable, as are any derived rules based on those theorems.

If, in doing a derivation, it is desirable to use a rule that has not yet been proved, it is possible to summon the menu (by using the Menu button), open a second instance of the Derivation module, prove the relevant theorem in the new instance of the derivation module, then save it and return to the original module to complete the original derivation with the use of the rule.

THE INFERENCE RULES OF CHAPTER III

Any discussion of the inference rules of Chapter III presupposes an understanding of when a given occurrence of a variable is bound to a given quantifier phrase. Test yourself by determining which occurrences of variables in the following formula (discussed in the text) are bound to which quantifier phrases. The answer is color coded in the footnote.¹

$$\forall y((\exists x(Fx \vee Hy) \vee Gx) \rightarrow \forall z(Fz \vee (Gy \rightarrow \forall xGz)))$$

Universal Instantiation

The inference rule Universal Instantiation (UI), which is represented within the program as

$$\forall xFx \therefore FA$$

is applicable only to universal generalizations, formulas which contain an initial universal quantifier phrase (something of the form “ $\forall x$ ”, where “ x ” stands for any variable) with scope over the whole remainder of the formula. The initial universal quantifier will be the primary connective of such a formula. The rule allows one to pick any *term* (in Chapter III, the terms consist of the variables, “ a ”, “ b ”, “ c ”, ... “ x ”, “ y ”, “ z ” and the Name Letters, “ A ”, ... “ E ”), and infer the *instance*, *for that term*, of the universal generalization. In our technical vocabulary, the variable that occurs in the quantifier phrase is called the *generalized variable*, and the term is called the *instantial term*.

Intuitively, if the universal generalization is read as saying that a certain formula is true of everything, the instance should say that the formula is true of the referent of the term. In the representation above, if we read “ $\forall xFx$ ” as saying that “ Fx ” is true of everything (true for *all* values of the variable “ x ”), then “ FA ” can be read as saying that “ Fx ” is true of A (the referent of “ A ”). There are two technical requirements that are provided in order to ensure that the result of applying UI *does* say of the referent of the term exactly what the universal generalization says about everything.

The first requirement is that *all and only* occurrences of the generalized variable that were bound by the relevant universal quantifier be replaced by the instancial term. (The text speaks of occurrences of the generalized variable that are free in the formula that immediately follows the quantifier phrase. This is equivalent to our formulation. The occurrences of the generalized variable that are bound by an initial quantifier phrase are exactly those that will become free when the quantifier phrase is removed.) The requirement that *all* such occurrences be replaced rules out such oddities as:

$$\forall x(Fx \rightarrow Gx \wedge Hx) \therefore (FA \rightarrow Gx \wedge HA)$$

in which, plainly, the conclusion does *not* say of A what the premise says of everything (namely, that if it has the property F , it has both of the properties G and H). The requirement that *only* such occurrences be replaced rules out such oddities as:

$$\forall x(Fx \rightarrow Gx \wedge \exists x(Hx \wedge \sim Gx)) \therefore (FA \rightarrow GA \wedge \exists x(HA \wedge \sim GA))$$

in which, again, the conclusion does *not* say of A what the premise says of everything (namely, that if it has the property F , it has the properties G and *something* has H). In the conclusion of the second inference, the quantifier phrase “ $\exists x$ ” is *vacuous*; it binds nothing. A clear example of why this inference

¹ $\forall y((\exists x(Fx \vee Hy) \vee Gx) \rightarrow \forall z(Fz \vee (Gy \rightarrow \forall xGz)))$ Note that the orange occurrence of “ x ” is free, and the green quantifier phrase is vacuous.

is fallacious can be seen by taking the scheme of abbreviation to be: Fx : x is a biological father, Gx : x is male, Hx : x is a biological mother.

The second requirement is that all occurrences of the generalized variable that were bound by the initial universal quantifier be replaced by the *free* occurrences of the instantial term. This rules out such oddities as:

$$\forall x(Fx \rightarrow \exists z(Gx \wedge Hz \wedge \sim Gz)) \therefore (Fz \rightarrow \exists z(Gz \wedge Hz \wedge \sim Gz))$$

in which the conclusion, again, does *not* say of z what the premise says of everything, namely, that if it_1 has the property F , then there is something such that it_1 has the properties G and that something, call it " it_2 ", has H and lacks G . Here we have had to resort to subscripted pronouns to do the work that the distinct generalized variables do. In the conclusion, the occurrence of " z " marked by the underline, which replaces an occurrence of " x " bound by the relevant universal quantifier, is not free as required. It has been "captured" (as we say) by the quantifier phrase " $\exists z$ ", and is bound to it. The conclusion can be read as saying that if z has the property F , then there is something such that has the property G and also has H and lacks G . The consequent of the conclusion is contradictory (nothing can both have and lack the property G), but there is nothing contradictory about the consequent of the premise (as reversion to the scheme of abbreviation given above will quickly show).

Existential Instantiation

The inference rule Existential Instantiation (EI), which is represented within the program as

$$\exists xFx \therefore Fa$$

is applicable only to formulas whose primary connective is an existential quantifier phrase (something of the form " $\exists x$ ", where " x " stands for any variable). The rule allows one to pick any *new variable* -- one that does not already occur, free or bound, on any prior line of the derivation (it can occur in a premise not yet brought into the derivation) -- as the instantial variable, and infer the *instance, for that variable*, of the existential generalization. The two technical requirements that apply to UI apply here too, namely that *all and only* occurrences of the generalized variable that were bound by the relevant universal quantifier be replaced by the instantial variable, and that all such replacements be by *free* occurrences of the instantial variable.

Intuitively, if the existential generalization is read as saying that a certain formula is true of at least one thing, the instantial variable may be thought of as an arbitrary name given to one of those things. The restriction to variables that have not already occurred is primarily (though not entirely) to avoid identifying representatives of groups that may have no members in common. (We call this an "external" restriction because it pertains to the whole derivation within which the rule is being applied, and not just to the relation between the premise and conclusion of the rule.) For example, given that there are students, we may wish to continue our reasoning by saying, "Let a be such a student." Now given that there are non-students, we may also say, "Let b be such a non-student." But it would plainly engender confusion if we were to call both the student and the non-student by the same name.

This explains why we cannot choose a variable already in use as a *free* variable, but the restriction is actually much stronger than that, it also restricts the use of any variable that has already appeared bound, even in an uncanceled Show line. This very strong restriction is necessary, in the presence of the rule for Universal Derivations, in order to avoid being able to derive a universal generalization from the corresponding existential generalization, for example to derive " $\forall yFy$ " from " $\exists xFx$ ".

Given the strong external restriction, it is good practice to reserve a little used range of variables for EI, say, those at the beginning of the alphabet, and EI first to " a ", next to " b ", next to " c ", etc. This is assuming that none of these has a prior occurrence in the derivation.

Existential Generalization

The inference rule Existential Generalization (EG), which is represented within the program as

$$FA \therefore \exists xFx$$

is applicable to any formula. In applying it, we must select the occurrences of the instancial term, here represented by “A”, on which we wish to generalize, and we must select the generalized variable, here represented by “x”, which will appear in the quantifier phrase. The two technical requirements that apply to UI apply here too, but *backwards*, namely from conclusion to premise rather than from premise to conclusion. This should not be surprising since in UI the generalized formula is the premise and the instance is the conclusion, whereas in EG the *instance* is the premise and the *generalized* formula is the conclusion. The requirement is that the premise be the result of replacing *all and only* occurrences of the generalized variable that are bound by the initial existential quantifier *in the conclusion* by free occurrences of the instancial term. Briefly, the relation between premise and conclusion is required to be such that the premise would follow from the conclusion by UI, if the existential quantifier were replaced by a universal quantifier.

Intuitively, if the existential generalization is read as saying that a certain formula is true of at least one thing, then it should follow from the premise that the formula is true of any given thing. The fact that the restriction runs backwards, from conclusion to premise, permits certain inferences that are intuitively correct, but may initially seem improper. For example, it is not required to generalize on *all* occurrences of the instancial term. The following are all correct applications of EG:

$$(FA \wedge GA) \therefore \exists x(Fx \wedge Gx)$$

$$(FA \wedge GA) \therefore \exists x(Fx \wedge GA)$$

$$(FA \wedge GA) \therefore \exists x(FA \wedge Gx)$$

$$(FA \wedge GA) \therefore \exists x(FA \wedge GA)$$

Intuitively, if Claire studies and passes, it follows that: someone studies and passes, someone studies and Claire passes, someone is such that Claire studies and the ‘someone’ passes, someone is such that Claire studies and passes (the vacuous case). So in applying EG, one must not only pick the instancial term, here “A”, one must decide *which occurrences of the instancial term* one wishes to generalize on. It need not be all of the occurrences.

Note that by applying the backwards UI test, all four inferences are seen to be correct:

$$\forall x(Fx \wedge Gx) \therefore (FA \wedge GA)$$

$$\forall x(Fx \wedge GA) \therefore (FA \wedge GA)$$

$$\forall x(FA \wedge Gx) \therefore (FA \wedge GA)$$

$$\forall x(FA \wedge GA) \therefore (FA \wedge GA)$$

The restriction on UI: that one cannot UI to a variable that becomes bound, which ruled out such attempts at UI as

$$\forall x(Fx \wedge \exists z(Gx \wedge Hz \wedge \sim Gz)) \therefore (Fz \wedge \exists z(Gz \wedge Hz \wedge \sim Gz))$$

as incorrect, appears here as a restriction on generalizing on a bound occurrence of the instantial term. This rules out

$$(Fz \wedge \exists z(\underline{Gz} \wedge Hz \wedge \sim Gz)) \therefore \exists x(Fx \wedge \exists z(Gx \wedge Hz \wedge \sim Gz))$$

in which the underlined, bound occurrence of “z” is generalized on.

The restriction on UI: that one must replace *all* bound occurrences of the generalized variable, which ruled out attempts at UI such as

$$\forall x(Fx \vee (Gx \wedge Hx)) \therefore (Fa \vee (Ga \wedge Ha))$$

as incorrect, appears here as a restriction on generalizing on two different instantial terms at once (or what is the same thing, generalizing on one term, but picking up a free occurrence of a variable different from the instantial term). This rules out

$$(Fa \vee (G\underline{x} \wedge Ha)) \therefore \exists x(Fx \vee (Gx \wedge Hx))$$

in which the underlined, free occurrence of “x” gets picked up (i.e. improperly bound).

Universal Derivation

Instead of having Universal Generalization as an inference rule analogous to EG, we have a fourth form of boxing and canceling known as “Universal Derivation” (UD). This form of boxing and canceling allows one to box and cancel a Show line of the form

Show $\forall xFx$

when one has obtained

Fx

(subject, of course, to the usual provisos that the displayed Show line is the last of the uncanceled Show lines and that the displayed formula is not boxed). In UD there is no replacing of occurrences of the generalized variable since the formula obtained must be exactly, symbol by symbol, the formula whose universal generalization appears in the Show line. Thus, it is *not* permitted to box and cancel if one obtained

Fy

instead of “ Fx ”.

There is an “external” restriction on the use of UD, as there is on EI. The generalized variable, here “x”, must not have occurred *free* in any line antecedent to the Show line that is to be canceled. Recall that an *antecedent* line is one which is neither boxed nor contains uncanceled Show. The restriction applies only to lines *above* the Show line to be canceled by UD. Note that this restriction is much less rigorous than that applying to EI. For EI, the variable may not have occurred in *any* earlier line. This includes bound occurrences in uncanceled Show lines. For UD, the variable is only restricted from *free* occurrences in *antecedent* lines.

The two restrictions work together to prevent there being a correct derivation for the argument,

$$\exists xFx \therefore \forall yFy$$

You may wish to construct derivations for this obviously invalid argument that would be correct if either of the two external restrictions were dropped, or even if the very strong restriction on EI were weakened to apply only to *antecedent* lines.

Quantifier Negation

There are two sets of very fundamental and useful derived rules: Quantifier Negation (QN) and Alphabetic Variance (AV). QN concerns the movement of a negation through a quantifier phrase, and thus is analogous to such laws as DM, NC, and NB. In essence, a negation can pass through a quantifier phrase if the phrase changes from universal to existential or vice versa. The laws are usually stated in a multiplicity of forms, some incorporating the elimination of double negations. The fundamental forms are:

$$\text{QN1} \quad \sim \forall x Fx \therefore \exists x \sim Fx$$

$$\text{QN3} \quad \sim \exists x Fx \therefore \forall x \sim Fx$$

Other forms are:

$$\text{QN2} \quad \exists x \sim Fx \therefore \sim \forall x Fx$$

$$\text{QN4} \quad \forall x \sim Fx \therefore \sim \exists x Fx$$

$$\text{QN5} \quad \forall x Fx \therefore \sim \exists x \sim Fx$$

$$\text{QN6} \quad \sim \exists x \sim Fx \therefore \exists x Fx$$

$$\text{QN7} \quad \exists x Fx \therefore \sim \forall x \sim Fx$$

$$\text{QN8} \quad \sim \forall x \sim Fx \therefore \exists x Fx$$

Alphabetic Variance

The second set of fundamental derived rules are the rules of Alphabetic Variance (AV), sometimes called “Rewrite of Bound Variable”. These laws allow the replacement of the generalized variable in an initial quantifier phrase with another provided that all variables remain bound to the same quantifier phrases. There are two such laws, one for the universal quantifier and one for the existential quantifier.

$$\text{AV1} \quad \forall x Fx \therefore \forall y Fy$$

$$\text{AV2} \quad \exists x Fx \therefore \exists y Fy$$